DEL MAR
EST. SEAFOODS 1988

NOTES FROM QINGDAO · 青岛会谈备忘

PLATE FREEZER PROJECT NOTES

平板冻结项目备忘

What we talked about and what we're after, so Wolfking can get started. English version; the Chinese version is a separate file.

DEL MAR SEAFOODS · 30 OCTOBER 2026 · FOR WOLFKING AND APCCO

WHO'S DOING WHAT

- **Del Mar (Joe)** — the customer. Picks the block size, approves the layout and the final design.
- **Wolfking** — designs and builds four vertical plate freezers with hydraulics and controls, plus the ammonia refrigeration package to run them. Sends drawings and the paperwork the permits need, and a couple of technicians to help APCCO's crew install and start up.
- **APCCO** — our local refrigeration contractor in California. Checks the design against California rules, stamps and files the permits, installs everything (piping, insulation, electrical) and services it afterwards. They don't build freezers; they're here to get it installed and legal. They'll talk straight to Wolfking's engineers and will ask a lot of questions. That's just the permit process.
- Let's keep everything on the one email thread with all three of us copied.

> **Sean, about the Poland story** (papers demanded at the port): before anything is built, APCCO will send Wolfking a list of the certificates and documents the permit will need, and Wolfking tells us which ones it can provide. No surprises at the port.

WHAT WE'RE MAKING

- **Whole California squid** in 10 kg blocks. A test for the China market: if Chinese buyers like plate-frozen squid, we convert more of our production. Squid season on Monterey Bay starts in May, so we want to be running by May 2027.
- **Ground salmon racks** (the frames left after filleting, ground to a paste) for a pet-food customer, year-round. That contract is what pays for the machines.

45 t	2 × 10 h	4	140
a day (50 short tons)	shifts + 4 h cleaning	vertical plate freezers	blocks of 10 kg per batch, per freezer (the wish)

Our math: 90 min freeze + 30 min unload/reload = a batch every 2 hours, 4 per shift, 8 a day. 45 t ÷ 8 ÷ 4 freezers ≈ 1.4 t, or 140 blocks, per freezer per batch. If the design lands a bit lower, that's okay.

Why we're doing this: today we blast-freeze squid in cartons on racks, which is a lot of hand labor at $30 an hour. These four freezers are a test. If it works, we add more of the same design, and later we automate loading, unloading, bagging and boxing.

THE BLOCK

- 10 kg, about 12 litres (12,000,000 mm^3). Squid is roughly 20% lighter than water, so 12 L ≈ 10 kg.
- **75 mm thick.** That one's fixed.

- Face around 535 × 300 mm. We're flexible on the exact face. Pick what fits your plates and still gives 12 L. Whatever we choose becomes our standard block for every freezer after this, so let's agree it before plates get ordered.
- We talked about two dividers per station, three blocks across. Check that works with your plate height.
- 90 minutes is our guess for 75 mm. Please check it and give us the real number for whole squid and for salmon paste.
- **Drainage, this one matters.** Whole squid dumps a lot of water and ink when you load it. The bottom of the station has to let that liquid out, so the customer gets a block of squid and not a block of ice. But the paste can't leak out. We need a bottom that handles both. Happy to send photos or samples.

THE FREEZERS

- Four identical vertical plate freezers, indoors, top-loaded. Hydraulic open/close and lift, hot-gas release, blocks come out one side onto a table or conveyor.
- Loading is manual or semi-automatic to start. We're thinking a conveyor with gates for whole squid and a hose/pump for the paste; your ideas are welcome. But build them so your automatic loading and unloading can bolt on later: clearance on top, mounting points, spare control I/O.
- Blocks go into a bag, then a box. You mentioned a partner's bagger and box sealer; quote it as an option.
- Materials: aluminum plates. Stainless or aluminum anywhere product or wash water touches, nothing galvanized above the plates. For the base frame, give us stainless and hot-dip galvanized with the price difference. It's a salt-water, chlorine-washdown plant.
- Hydraulic unit: 460 V / 60 Hz motor, placed out of the cold and wet area, happy with food-grade (H1) oil, rated for −35 °C at the cylinders. Tell us if it's one unit per freezer or one shared.
- Stainless, ammonia-rated flex hoses that can be swapped from outside. No copper or brass anywhere near the ammonia.
- E-stops, guarding at the open top, a lockable disconnect. APCCO will check it against US rules.
- English manuals, drawings, electrical schematics, a two-year spares list, and STEP models so we can lay out the room.

REFRIGERATION

- **Ammonia (R717).** California basically rules out Freon at this size.
- Wolfking supplies the whole package: compressors, condenser, vessels, pumps, valves, controls. APCCO runs the piping and wiring on site.
- We like the **containerized engine room** idea: outside, next to the building, freezers inside, insulated pipes between. Saves space and keeps the noise outside. Joe and APCCO are meeting at the plant in early November to pick the spot, so send your standard container dimensions before then. Joe also asked about a small self-contained unit on each freezer; tell us if that's realistic with ammonia and what it would cost.

- **Keep the ammonia charge as low as you can.** In California, 500 lb (227 kg) on site is where the extra paperwork (CalARP) starts. If we can't get under that at this size, fine, just tell us the number.
- APCCO's rough early estimate, before we went to 140 blocks: about 45 TR at −30 °F (−34 °C) suction, pumped recirculation. Please do your own load calc for the final batch size.
- Tell us: system type and recirculation ratio; suction and condensing temperatures; condenser type (evaporative or air-cooled; water and noise both matter here); receiver or not; defrost method (hot gas?) and what the freezers need for it; which valves come mounted, which come loose, which you expect APCCO to supply, with brands.
- Vessels need the ASME U-stamp, piping to ASME B31.5, and the plates have to be rated for discharge pressure because hot gas hits them. APCCO will give you the numbers. California wants weld X-rays on some vessels; APCCO will say which.

POWER

- 480 V, 3-phase, 60 Hz. Motors nameplated 460 V / 60 Hz. There's no 380 V and no 220 V single-phase in the US.
- Controls on 120 VAC and 24 VDC.
- Panels built to UL 508A (UL or ETL listed), NEMA 4X stainless where there's washdown, name-brand UL components (Schneider, ABB, Siemens, Eaton, Allen-Bradley).
- Motors NEMA Premium / IE3. The US requires it for imports, even inside a machine. Size pumps and fans for 60 Hz speed, not relabelled 50 Hz units.
- **Send a load list early:** kW, amps and starting current per freezer and per skid. We need it to apply to the power company, and that can take longer than building the machines.
- PLC and screens in English. Remote access for your support is fine. Siemens or Allen-Bradley is easiest to service here.

THE SITE

- Watsonville, California, on Monterey Bay, about 150 km south of San Francisco. Mild coastal weather; APCCO will send design temperatures for the condenser.
- Earthquake country (Loma Prieta 1989 was right here). Everything gets anchored. Send weights, centres of gravity and anchor-bolt locations; APCCO's engineer does the calculations. Footprints and floor loads too.
- Ship to Oakland or Long Beach, then truck to Watsonville. Keep the engine room package within container or road-legal size.

CALIFORNIA THINGS TO KNOW WHILE YOU DESIGN

The short version. APCCO is the authority on all of this and will give the exact values.

- Permits go through the City of Watsonville on drawings stamped by APCCO's California engineer. Permits can take longer than manufacturing, so drawings come first.

- Ammonia design to IIAR 2 (and IIAR 4/5/6 for install and start-up) plus the California Mechanical Code: design pressures, relief valves, valve labelling, emergency shutdown, gas detection and ventilation for indoor machinery rooms. An outdoor package avoids most of the indoor part.
- 500 lb of ammonia triggers CalARP; 10,000 lb triggers federal PSM/RMP.
- ASME U-stamp vessels, B31.5 piping, Cal/OSHA vessel permits with weld radiography where required. Material certs and welder qualifications on file.
- Every electrical item needs a US listing (UL, ETL or CSA). Unlisted gear means a field evaluation, which is slow and expensive.
- Seismic anchorage per the California Building Code.
- It's a food plant under FDA rules: cleanable, corrosion-resistant surfaces, H1 lubricants, no flaking coatings over product.
- Cal/OSHA guarding, E-stops and lockout. Noise limits at the property line for outdoor gear.

TIMING

- **Now:** these notes; Joe and APCCO meet at the plant on the layout; Wolfking starts.
- **As soon as you can:** preliminary drawings, P&ID, load list, ammonia charge estimate and a proposal. The permit work starts from the preliminary design.
- **Build:** about 100 days after design approval, per your estimate at the show.
- **Running by May 2027,** June at the latest. Joe told the pet-food customer six months.
- **2027–28:** automatic loading and unloading, bagging, boxing, check-weighing.

WHAT WOULD HELP US MOST RIGHT NOW

1. A preliminary layout of one freezer: size, plate count, your block-face proposal, loading height, discharge side, weight.
2. A freeze-time check for 75 mm whole squid and salmon paste.
3. The refrigeration concept: system type, P&ID, equipment list, ammonia charge, container dimensions.
4. Electrical load list and one-line.
5. Vessel list with ASME status, and which certificates you can supply.
6. A proposal: what's Wolfking's and what's APCCO's, price, delivery, terms, spares, technician terms.
7. STEP models when you have them.
8. A couple of US references we can call (Oregon, LA).
9. A ballpark for the phase-2 automation and the bagger/boxer, so we design the interfaces now.

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